

[0] $a^n + b^n = c^n$ where $a < b < c$ and a, b, c , and n are positive integers with $n \geq 2$
 since $c > a$ and $c > b$, c can be written as $x+a$ or $y+b$ where x and y are positive integers

[1] thus $a^n + b^n = (x+a)^n = (y+b)^n$

now expand $(x+a)^n$ and $(y+b)^n$:

$(x+a)^n = x^n + a^n + \text{remaining terms (call their sum "f")}$

$(y+b)^n = y^n + b^n + \text{remaining terms (call their sum "g")}$

thus $a^n + b^n = x^n + a^n + f = y^n + b^n + g$

subtract from both sides

$b^n = x^n + f$

$a^n = y^n + g$

f and g are both positive integers, thus $b^n > x^n$ and $a^n > y^n$

take n th root of both sides

$b > x$ and $a > y$

write b as $x+s$ and a as $y+r$ where s and r are positive integers

now back to [1]

$a^n + b^n = (x+a)^n = (y+b)^n$

$a^n + b^n = (x+(y+r))^n = (y+(x+s))^n$

$a^n + b^n = (x+y+r)^n = (x+y+s)^n$

thus $r = s$; rename this integer Z .

[2] then $a^n + b^n = (x+y+Z)^n$

by definition, $a \neq b$ and since $c = x+a = y+b$, $x \neq y$

[3] $y+Z = a$ and $y < a$, thus **a must be at least 2** (if $y = Z = 1$)

but if $y = 1$ and $x \neq y$, then x must be at least 2.

$b > a$ so **b must be at least 3**

[4] thus $a + b \geq 5$

now back to [2]:

$a^n + b^n = (x+y+Z)^n$

recall that $b = x+Z$ and $a = y+Z$. thus $x = b-Z$ and $y = a-Z$. Substitute these values:

$a^n + b^n = ((b-Z)+(a-Z)+Z)^n$

[5] $a^n + b^n = (a + b - 2Z + Z)^n = (a + b - Z)^n$

Substituting the minimum values of a, b , and Z into [5] gives

$2^n + 3^n = (2 + 3 - 1)^n = 4^n$

Divide both sides by 4^n

[6] $(1/2)^n + (3/4)^n = 1$

Both terms on the left are less than 1 and get smaller as n increases, thus n must be small.

Substituting $n = 1$ gives $(1/2) + (3/4) = 1.25$ which is greater than 1

Substituting $n = 2$ gives $(1/4) + (9/16) = 0.8125$ which is less than 1

All larger values of n will give a value for $[6]$ that is less than 0.8125, since both exponent terms are decreasing

Thus n must be between 1 and 2 if $a = 2$, $b = 3$, and $Z = 1$. But this violates our initial condition that n is a positive integer

Thus $a = 2$, $b = 3$, and $z = 1$ cannot be correct

[7] The next lowest possible value for a is 3. Let $a = 3$, $b = 4$, and $Z = 1$. Then we have

$$3^n + 4^n = (3 + 4 - 1)^n = (2 * 3)^n$$

$$3^n + 4^n = (2^n)(3^n)$$

$$1 + (4/3)^n = 2^n$$

$$2^n - (4/3)^n = 1$$

For $n = 1$ we have $2 - 4/3 = 2/3 \neq 1$, so n cannot be 1

For $n = 2$ we have $2^2 - (4/3)^2 = 4 - (16/9) = (36/9) - (16/9) = 20/9 = 2.222... \neq 1$, so n cannot be 2

But we know n CAN be 2, since $3^2 + 4^2 = 5^2$

Thus the problem must be with the minimum value of Z , not those of a or b

[8] So **Z must be at least 2**

[9] and **$a + b \geq 7$**

Now return to [5].

$$a^n + b^n \leq (a + b - Z)^n$$

Z must be at least 2.

$$a^n + b^n \leq (a + b - 2)^n$$

We know the minimum value of the left side is $3^2 + 4^2 = 5^2 = 25$.

[10] Thus $25 \leq a^n + b^n \leq (a + b - 2)^n$

Substitute $k = a+b$

$$25 \leq (k-b)^n + (k-a)^n \leq (k-2)^n$$

Since $b > a$, we know $(k-b)^n < (k-a)^n$. Thus we can write

$$25 \leq 2^*(k-b)^n \leq (k-b)^n + (k-a)^n \leq (k-2)^n$$

Or simply

$$[11] 25 \leq 2^*(k-b)^n \leq (k-2)^n$$

Write the factor 2 (from the middle part) as $(2^{1/n})^n$

$$25 \leq (2^{1/n})^n * (k-b)^n \leq (k-2)^n$$

Take the n th root of both sides

$$25^{(1/n)} \leq 2^{(1-n)^*(k-b)} \leq k - 2$$

$$(25^{(1/n)})/(k - b) \leq 2^{(1-n)} \leq (k - 2) / (k - b)$$

$$(25^{(1/n)})/a \leq 2^{(1-n)} \leq (a + b - 2) / a$$

$$(25^{(1/n)})/a \leq 2^{(1-n)} \leq 1 + (b/a) - (2/a)$$

The largest possible value of $(25^{(1/n)})/a$ is at $a = 3$. So

$$(25^{(1/n)})/a \leq (25^{(1/n)})/3 \leq 2^{(1-n)} \leq 1 + (b/a) - (2/a)$$

Ignore the rightmost part.

$$(25^{(1/n)})/a \leq (25^{(1/n)})/3 \leq 2^{(1-n)}$$

Raise everything to the power of n .

$$25 / (a^n) \leq 25 / (3^n) \leq 2$$

Ignore the leftmost part.

$$25 / (3^n) \leq 2$$

$$3^n / 25 \leq 1/2$$

$$3^n \leq 25/2$$

$$n \leq \log_{\text{base } 3} \text{ of } 12.5$$

$$n \leq 2.2990\dots$$

The only possible integer value of n is 2.

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